

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical
Practical – VII

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Class: TYIT	Batch:01
Date of Assignment: 30-07-2026	Date/Time of Submission: 30-07-2026

a. Develop a PWA that works in offline mode using Service Worker.

Code:

#Create wwwroot/manifest.json:-

```
{
  "name": "HTML TO PWA",
  "short_name": "HTP",
  "start_url": "/",
  "display": "standalone",
  "icons": [
    {
      "src": "icon/icon2.png",
      "sizes": "447X447",
      "type": "image/png",
      "purpose": "any"
    },
    {
      "src": "icon/icon3.png",
      "sizes": "447X447",
      "type": "image/png",
      "purpose": "any"
    }
  ]
}
```

#Create wwwroot/service-worker.js:-

```
self.addEventListener("install", e => {
  console.log("Service workers Installed");
});
self.addEventListener("fetch", e => {
  e.responseWith(fetch(e.request));
});
```

#Add to _Layout.cshtml

#Inside the <head> section:

```
<link rel="manifest" href="/manifest.json"/>
```

#Before the closing </body> tag:

```
<script>
  if('serviceWorker' in navigator){
    navigator.serviceWorker.register('/service-worker.js');
  }
</script>
```

offline.html:-

```
<!DOCTYPE html>
<html>
<head>
  <title>Offline</title>
</head>
<body>
```

```

<header>
  <h2>My College Website</h2>
</header>
<main>
  <h2> You are Offline</h2>
  <p>Please check your internet connection.</p>
  <button onclick="location.reload()">Try Again</button>
</main>
<footer>
  <p>© 2026 My College Website</p>
</footer>
</body>
</html>

```

Output:-



b. Create a Student Record Management PWA.

Code:

#Create wwwroot/manifest.json:-

```

{
  "name": "HTML TO PWA",
  "short_name": "HTP",
  "start_url": "/",
  "display": "standalone",
  "icons": [
    {
      "src": "icon/icon2.png",
      "sizes": "447X447",
      "type": "image/png",
      "purpose": "any"
    },
    {
      "src": "icon/icon3.png",
      "sizes": "447X447",
      "type": "image/png",
      "purpose": "any"
    }
  ]
}

```

#Create wwwroot/service-worker.js:-

```

const CACHE = "v1";
self.addEventListener("install", e => {
  e.waitUntil(
    caches.open(CACHE).then(cache => {

```

```

        return cache.addAll([
            "/",
            "/Home/Index",
            "/css/site.css",
            "/js/site.js"
        ]);
    });
});
self.addEventListener("fetch", e => {
    e.respondWith(
        caches.match(e.request).then(response => {
            return response || fetch(e.request);
        })
    );
});

```

#Add to _Layout.cshtml

#Inside the <head> section:

```
<link rel="manifest" href="/manifest.json"/>
```

#Before the closing </body> tag:

```

<script>
    if('serviceWorker' in navigator){
        navigator.serviceWorker.register('/service-worker.js');
    }
</script>

```

offline.html:-

```

<!DOCTYPE html>
<html>
<head>
    <meta charset="utf-8" />
    <title></title>
</head>
<body>
    <h2>offline</h2>
</body>
</html>

```

HomeController.cs:-

```

public IActionResult Index()
{
    Student S = new Student
    {
        RollNo = 6,
        Name = "Nandini"
    };
    return View(S);
}

```

Student.cs:-

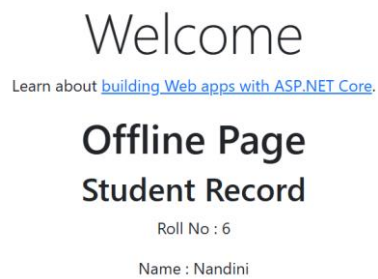
```

namespace pract7.Models
{
    public class Student
    {
        public int RollNo { get; set; }
        public required string Name { get; set; }
    }
}

```

index.cshtml:-

```
@{
    ViewData["Title"] = "Home Page";
}
<div class="text-center">
    <h1 class="display-4">Welcome</h1>
    <p>Learn about <a href="https://learn.microsoft.com/aspnet/core">building Web apps with
ASP.NET Core</a>.</p>
    <h1>Offline Page</h1>
    <h2>Student Record</h2>
    <p>Roll No : @Model.RollNo</p>
    <p>Name : @Model.Name</p>
</div>
```

Output:

Welcome

Learn about [building Web apps with ASP.NET Core](https://learn.microsoft.com/aspnet/core).

Offline Page

Student Record

Roll No : 6

Name : Nandini

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Welcome

Learn about [building Web apps with ASP.NET Core](https://learn.microsoft.com/aspnet/core).

Offline Page**Student Record**

Roll No : 6

Name : Nandini

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